

PURNASEKHAR SUNKARA

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Technical Skills

Languages: Python, TypeScript, JavaScript, SQL

Full Stack: Node.js, React.js, Next.js, FastAPI, REST APIs, HTML5, CSS3, Tailwind CSS

AI & LLM: OpenAI API, Anthropic API, Google Gemini, LangChain, LlamaIndex, Prompt Engineering, RAG Pipelines, Function Calling, Tool Use, Agent Workflows, Transformer Architecture, Cursor, GitHub Copilot, ChatGPT, Claude

Vector Databases: FAISS, ChromaDB, Pinecone (familiar), Weaviate (familiar), Embedding Pipelines, Semantic Search

Backend & Data: Supabase, PostgreSQL, MongoDB, Redis, Data Modeling, Async Programming, OOP

Cloud & DevOps: AWS (EC2, S3, RDS), Docker, GitHub Actions, CI/CD, Git, Vercel, Linux/Unix

Projects

SuperClaims - LLM Multi-Agent System | *Python, FastAPI, OpenAI API, LanfChain, Docker*

[Github](#)

- Built production LLM agent system integrating OpenAI API with function calling and tool-use patterns implementing few-shot prompting, chain-of-thought reasoning, and system prompt engineering throughout. Designed RAG pipeline with vector search and semantic retrieval built evaluation loops measuring output quality with regression testing and iterative prompt improvements. Applied transformer architecture understanding to optimize LLM integration managing context windows, attention patterns, and structured output generation reliably.

SmartPDF Chat - RAG Pipeline | *Python, LlamaIndex, FAISS, Google Gemini, FastAPI*

[Github](#)

- Built end-to-end RAG pipeline document ingestion, chunking, embedding generation, FAISS vector indexing, semantic retrieval, reranking, and Gemini AI response generation. Implemented vector database search with ChromaDB/FAISS reducing hallucination rates by 30% through retrieval quality evaluation and moderation hooks. Applied prompt engineering and context management maintaining conversation state and delivering consistent LLM responses across production workflows.

AEO Diagnostic Tool | *React, Next.js, TypeScript, Node.js, Supabase*

[Github](#)

- Built full stack applications using React, Next.js, and TypeScript integrating Supabase authentication, real-time database, and storage with REST APIs end-to-end. Deployed on AWS and Vercel with complete Git workflow optimizing performance, scalability, and reliability across production environments. Integrated LLM APIs for AI-powered features leveraging OpenAI and Anthropic APIs to deliver intelligent, data-driven user experiences throughout.

Experience

Full Stack & Generative AI Engineer Intern

Mar 2024 – Jan 2026

EPAM Systems · DataValley IT Solutions · Zidio Development · Future Intern

Remote

- Built and maintained full stack applications using TypeScript, Node.js, React, and Next.js developing responsive frontend interfaces and scalable backend services with clean, production-ready code.
- Integrated LLM APIs including OpenAI, Anthropic, and Google Gemini building AI-powered features, agent workflows, and tool-use patterns with prompt engineering and function calling throughout.
- Built RAG pipelines with vector database integration implementing document ingestion, embedding generation, semantic retrieval, and reranking for intelligent knowledge retrieval systems.
- Applied prompt engineering techniques including few-shot, chain-of-thought, and system prompts optimizing LLM outputs for consistency, accuracy, and production reliability.
- Built LangChain and LlamaIndex orchestration workflows designing agent-based systems with tool use, context management, and multi-step reasoning across production environments.
- Integrated Supabase for authentication, database, and storage needs deploying and managing full stack applications on AWS with complete production ownership.
- Worked with vector databases including FAISS and ChromaDB implementing semantic search, similarity retrieval, and embedding pipelines for AI-driven use cases.
- Optimized application performance, scalability, and reliability debugging issues, writing clean code, and following best engineering practices throughout.

Education

BVC Engineering College

July 2020 - July 2024

Bachelor of Technology in Computer Science & Engineering, GPA : 8.42

Certifications

- **Full Stack Developer Certification** - DataValley IT Solutions
- **Front-End Technologies Certification** - EPAM Systems
- **Azure Fundamentals Certification** - Microsoft
- **Fundamentals of Cloud Security** - Palo Alto Networks

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